

Northern Red Oak

Quercus rubra

Species background: Native to eastern North America; one of the most commercially important hardwood trees on the continent and the state tree of New Jersey. Its leaf lobes end in sharp bristle-tips rather than rounded points -- a defining trait of the entire red oak group, distinct from the rounded lobes of white oaks like swamp white oak.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Northern Red Oak: Identify red oak by its pointy leaf lobes and small-capped acorn.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

My leaves have pointy, spiky tips -- that's how you know I'm in the 'red oak' family. Look under me in fall for acorns wearing tiny berets!

Procedure

- 1 Engage: Ask students to find something pointy nearby.
- 2 Explore: Look at a real or pictured leaf and count the pointy lobes.
- 3 Explain: Explain that pointy lobes are the sign of a 'red oak.'
- 4 Art: Draw an acorn wearing its cap like a tiny hat, then give it a silly face.
- 5 Evaluate: Student can point out the pointy lobes and describe the acorn's small cap.

Discussion questions

What animals might eat this tree's acorns?

Assessment

Correctly identifies pointy lobes as the red oak clue.

Extension

Look for acorns and their caps on the ground beneath the tree.

Northern Red Oak: Compare red oak's acorn and leaf to another oak species.

Standards connection: NGSS 3-LS4-3; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Find an acorn and compare its cap to a pin oak's -- bigger or smaller? Its acorns are an important fall food source for squirrels, blue jays, deer, and wild turkeys.

Procedure

- 1 Engage: Ask which animals students think eat acorns.
- 2 Explore: Compare an acorn's cap size to another oak's if available.
- 3 Explain: Discuss the acorn's role as a key fall food source for several animals.
- 4 Art: Draw an acorn wearing its little cap like a hat, then invent a tiny character who lives inside it.
- 5 Evaluate: Student names at least two animals that eat this tree's acorns.

Discussion questions

What might happen to those animals in a year when the oak makes very few acorns?

Assessment

Names two or more correct acorn-eating animals.

Extension

Count how many acorns you can find under the tree in five minutes.

Northern Red Oak: Explain the red oak vs. white oak grouping as a real field-ID dichotomy.

Standards connection: NGSS MS-LS4-2; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, graphite pencils.

Teacher background

Red oaks have bristle-tipped leaves and acorns that take two years to mature; white oaks have rounded lobes and one-year acorns -- a real dichotomy botanists use for field ID.

Procedure

- 1 Engage: Ask how scientists might sort dozens of oak species into just two useful groups.
- 2 Explore: Compare a red oak leaf's bristle-tipped lobes to a white oak's rounded lobes (real or pictured).
- 3 Explain: Introduce the red oak/white oak dichotomy and the one-year vs. two-year acorn timeline.
- 4 Art: Create a labeled botanical sketch comparing a red oak leaf (pointed, bristle-tipped lobes) to a white oak leaf (rounded lobes) side by side.
- 5 Evaluate: Write a paragraph explaining the red oak/white oak grouping using both leaf shape and acorn timing as evidence.

Discussion questions

Why might grouping species by a simple, observable trait like leaf-tip shape be more useful in the field than a genetic test?

Assessment

Paragraph correctly uses both leaf shape and acorn timing as evidence.

Extension

Find a white oak group species (like swamp white oak) nearby and compare it directly.

Northern Red Oak: Connect acorn maturation timing to squirrel caching behavior and oak reproductive strategy.

Standards connection: NGSS HS-LS4-1; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, graphite pencils, stippling tools.

Teacher background

Red oaks take two years to mature acorns while white oaks take one; research how this difference relates to the deeper evolutionary split between the two oak subgenera, and explain why acorn maturation time affects a squirrel's caching -- and accidental planting -- behavior differently for each group.

Procedure

- 1 Engage: Would a squirrel treat a one-year acorn and a two-year acorn the same way when deciding whether to eat it now or bury it?
- 2 Explore: Research documented differences in squirrel caching behavior between red oak and white oak acorns.
- 3 Explain: Connect maturation timing, tannin content, and caching behavior to the oaks' reproductive success.
- 4 Art: Produce a detailed graphite illustration of the acorn and cap using stippling to show texture, following the conventions of 19th-century natural history illustration.
- 5 Evaluate: Submit a written explanation of the caching-behavior evidence plus the stippled illustration.

Discussion questions

Is a squirrel 'helping' the oak reproduce, or is the oak's acorn chemistry manipulating the squirrel's behavior -- or both?

Assessment

Rubric covering evidence quality in the written explanation and illustration technique.

Extension

Research the acorn-tannin content difference between red and white oak groups and its effect on caching decisions.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.